

Introduction

Scoring goals is the most important statistic when it comes to determining the outcomes of a football match. It is also used in identifying what factors contribute to a player's performance score in informing team tactics, player scouting and analysis. The recent FIFA World Cup held in USA, Canada and Mexico has over 48 national teams participating that generates a diverse and multifaceted data by the player pool than any other prior World Cup. In this project we are aiming to infer two related research questions using different statistical models taught in this course.

Research Question 1: **What factors contribute to the number of goals a player scores?**

Research Question 2: **What factors contribute to whether a player scored during the tournament?**

Where Question 2 is aggregated based on Question 1, we transform the goals counts into a score/did-not score binary response. By fitting both multilinear regression on the count and logistic regression on binary outcomes of the same underlying quantity, we can check whether these two approaches would lead to the same predictors used in modeling and the same statistical conclusion.

EDA

Variables and Analysis Sample

The dataset consists of thirteen variables with two are categorical and eleven numeric. All the numeric performance measures are totals for the whole tournament, not per-match rates.

- **team country** — the national team the player played for. There are 48 teams, with 17 to 25 players each.
- **position** — playing position. Some players are listed with two roles, such as "FW, MF".
- **age** in years at the tournament. (year)
- **minutes** — total minutes played. (min)
- **goals** — total goals scored. (count)
- **assists** — passes that led directly to a team-mate scoring. (count)
- **cards_yellow** — yellow cards received. (count)
- **shots** — total shot attempts. (count)
- **shots_on_target** — shots that forced a save or went in. (count)
- **fouls** — fouls committed. (count)
- **offsides** — times the player was caught offside. (count)

- **interceptions** — opposition passes intercepted. (count)
- **tackles_won** — tackles won. (count)

The file has 1,248 rows. In 209 of them (16.7%) every match statistic is missing at once, including minutes. These are squad members who never played, so there is nothing to observe rather than data lost at random. We dropped them, which leaves 1,039 players with at least one minute. Nothing else is missing, so the sample is complete on all thirteen variables. We also kept the first role listed in position as the player's main position, giving MF (392), DF (349), FW (236) and GK (62).

All 62 goalkeepers have zero goals and zero shots. That is true by the nature of the position rather than something we measured. [State your decision here: drop them, leaving 977 outfield players, or keep them and note that the GK coefficient is a fixed zero.]

Summary Statistics

Variable	Mean	SD	Min	Q1	Median	Q3	Max	Skew	% zero
age	27.464	4.198	17	24	27	30	41	0.310	0.0
minutes	203.474	158.361	1	71	180	288	810	0.871	0.0
goals	0.284	0.835	0	0	0	0	10	5.223	82.8
assists	0.216	0.590	0	0	0	0	7	4.240	83.9
shots	2.481	3.800	0	0	1	3	41	3.654	34.6
shots_on_target	0.853	1.808	0	0	0	1	23	4.997	61.9
fouls	2.321	2.489	0	0	2	4	15	1.507	27.9
cards_yellow	0.260	0.512	0	0	0	0	3	1.886	77.4
offsides	0.354	0.886	0	0	0	0	9	3.976	78.9
interceptions	1.674	2.200	0	0	1	2	15	1.877	41.2
tackles_won	1.949	2.493	0	0	1	3	20	2.156	35.3

Table 1. Summary statistics for the 1,039 players in the analysis sample. The last column gives the percentage of players with a value of zero.

There are two findings worth mentioning: Every performance variable has a standard deviation bigger than its mean, and for seven of the nine the third quartile is 0 or 1. These are highly skewed counts that most players record nothing and a few record a lot; the skewness of our data runs from 1.5 for fouls up to 5.2 for goals. Age and minutes are the only variables that behave like ordinary continuous measurements. Age is close to symmetric around 27 and is the only variable in the data that looks roughly normal. Minutes has a mean of 203.5, about two and a quarter full matches, and is

moderately **right-skewed**.

The Response Variable

Plot (a) of Figure 1 shows the goal counts. Most of the mass sits at zero: 860 of 1,039 players (82.8%) did not score, 119 scored once, and only 60 scored two or more, with a maximum of 10. The variance (0.697) is 2.45 times the mean (0.284), so the counts are and clearly overdispersal.

Plot (b) shows the same thing after we collapse it to the binary response, where 179 players (17.2%) scored at least once. This throws away the difference between a one-goal player and a ten-goal player, but it turns an awkward count into something logistic regression handles well. That trade-off is the reason for comparing the two models.

Plots (c) and *(d)* show the two continuous predictors. Minutes is right-skewed with a lot of players under 200 minutes, which makes sense given how many teams go out in the group stage and how often substitutes are used. Age is symmetric and unimodal.

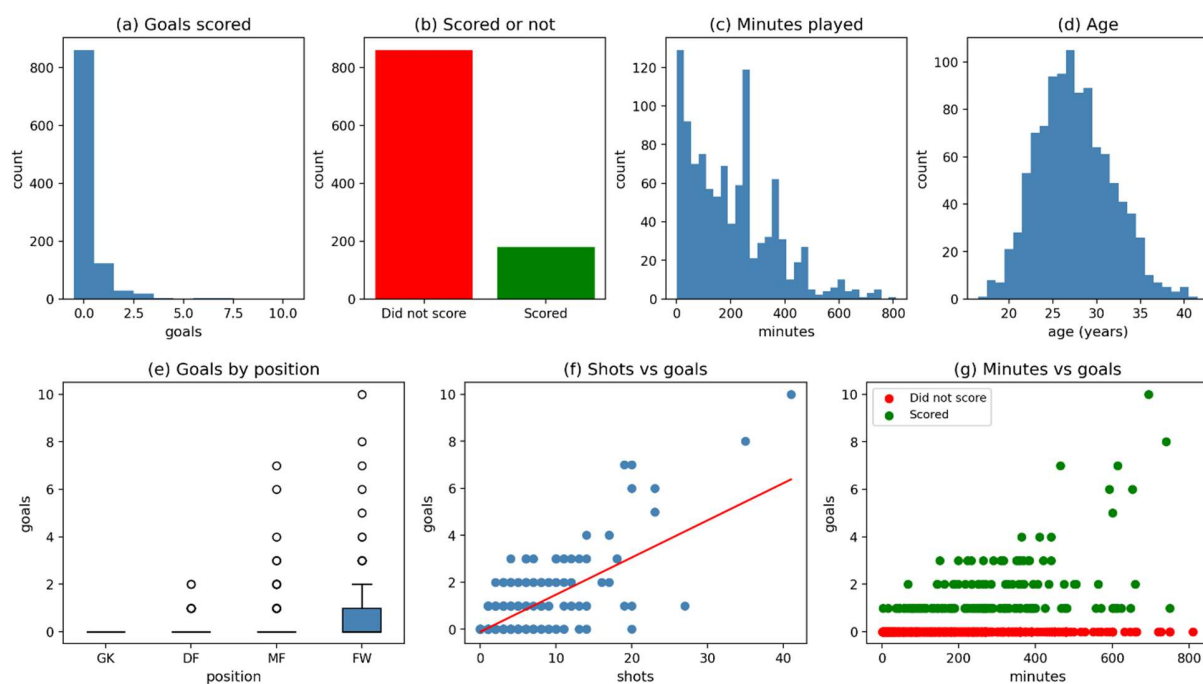


Figure 1. Exploratory plots for the 1,039 players with at least one minute. (a) goals; (b) the binary scored / did-not-score response; (c) minutes played; (d) age; (e) goals by position; (f) shots against goals with a fitted line; (g) minutes against goals, coloured by whether the player scored.

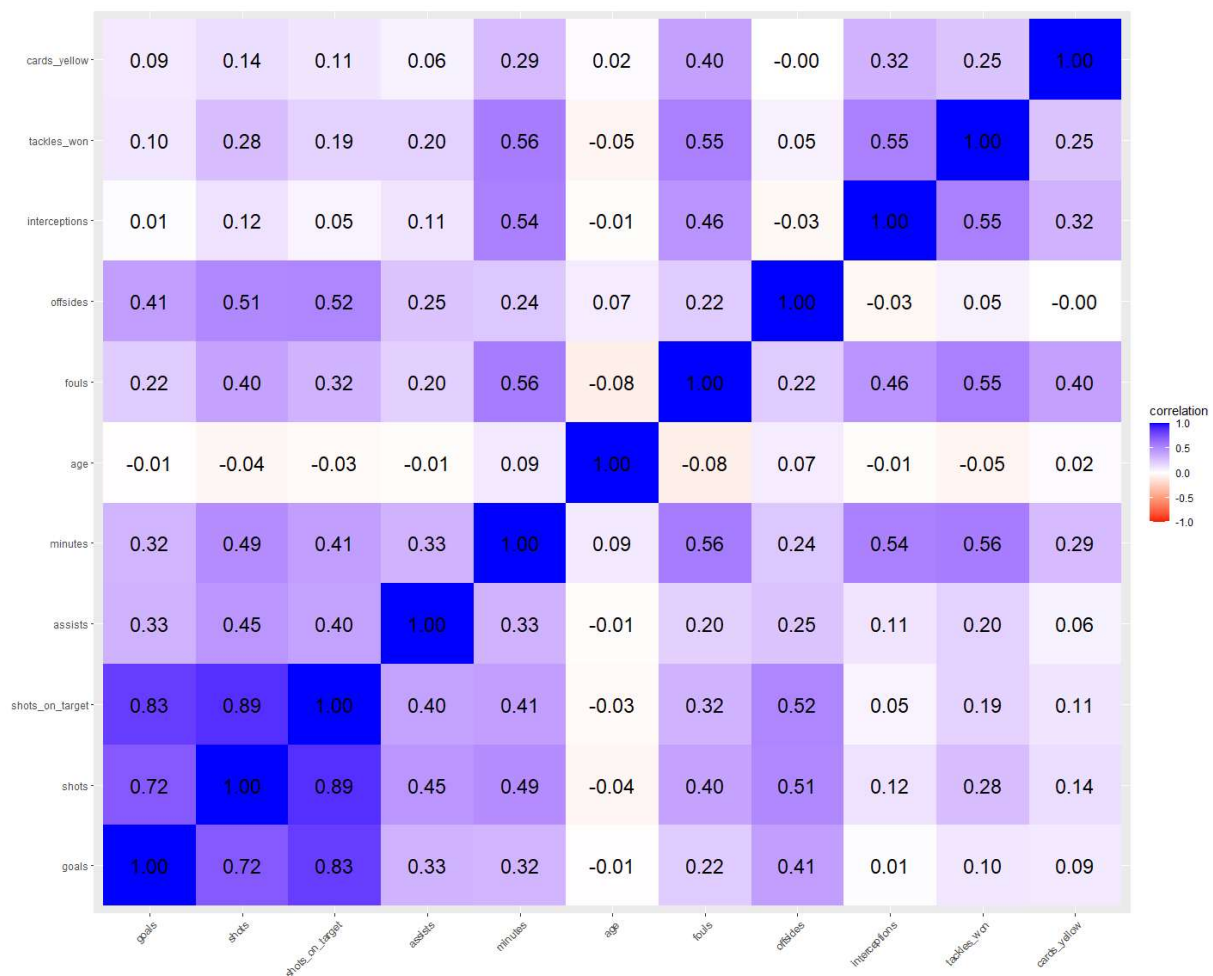


Figure 2. Correlation matrix for the eleven numeric variables ($n = 1,039$).

Differences by Position

Table 2 and Plot (e) show that **scoring depends heavily on position**. Mean goals go from 0.000 for goalkeepers to 0.089 for defenders, 0.329 for midfielders and 0.572 for forwards, and the share who scored at least once follows the same order. Forwards did this on the fewest minutes of any outfield position (139.1 against 220.5 for defenders), so role appears to matter more than time on the pitch. Position should be in both models.

Position	n	Mean goals	Mean shots	Mean minutes	% scoring ≥ 1
GK	62	0.000	0.00	310.6	0.0%
DF	349	0.089	1.21	220.5	8.0%

Position	n	Mean goals	Mean shots	Mean minutes	% scoring ≥ 1
MF	392	0.329	3.32	210.1	23.0%
FW	236	0.572	3.63	139.1	25.8%

Table 2. Scoring by main position ($n = 1,039$).

Predictors and Collinearity

Plots (f) and (g) plot goals against the two obvious predictors. **Shots is positively related to goals ($r = 0.72$) and shots on target more so ($r = 0.83$).** Minutes is only weakly related ($r = 0.32$). Plot (g) shows that the players who scored are spread across the whole range of minutes, while a large group with high minutes, including every goalkeeper and most defenders, scored nothing. Time on the pitch on its own says little about attacking output. Of the rest, offsides ($r = 0.41$) and assists ($r = 0.33$) are moderately related to goals, which fits since both go with attacking play. Fouls ($r = 0.22$), tackles won ($r = 0.10$), yellow cards ($r = 0.09$) and interceptions ($r = 0.01$) are weak, and we expect them to add little once position is in the model.

Figure 2. gives the full correlation matrix. The main problem is that it reveals is that shots and shots on target correlate at 0.89. They measure much the same thing, so putting both in a model needs a variance inflation factor check first. Minutes is also correlated with fouls (0.56), tackles won (0.56) and interceptions (0.54), which is expected for counts that build up over playing time.

Summary of Findings

Four important points from the EDA carry into the modelling.

- **The sample needed one exclusion.** 209 of the 1,248 records are players who never took the field and have no match statistics at all. Dropping them leaves a complete sample of 1,039.
- **The response is mostly zeros and dispersal.** 82.8% of players did not score and the variance of goals is 2.45 times its mean. Most other performance variables are skewed the same way, with skewness up to 5.2.
- **Position matters most, more than playing time.** Mean goals rise steadily from goalkeepers to forwards even though forwards played the fewest minutes of any outfield position, and minutes on its own correlates with goals at only 0.32.
- **The shooting variables are the strongest predictors but overlap.** Shots on target ($r = 0.83$) and shots ($r = 0.72$) lead, then offsides ($r = 0.41$) and assists ($r = 0.33$). Shots and shots on target correlate at 0.89, so we should keep one of them or combine them into a measure of accuracy.

These points set up the comparison this project is built around. Fitting a linear regression to a response that is 82.8% zeros will likely break the normality and constant

variance assumptions and can give negative fitted values. The logistic model avoids that but answers a coarser question. The next section fits both and checks whether they point to the same predictors.

Appendix

```
library(tidyverse)
library(patchwork)

wc <- read_csv("players-selected-columns.csv")

#Data Cleaning

wc <- wc %>%
  mutate(pos = factor(sub(".*", "", position), levels = c("GK", "DF", "MF", "FW")))

d <- wc %>%
  filter(!is.na(minutes)) %>%
  mutate(scored = factor(if_else(goals > 0, "Scored", "Did not score"),
    levels = c("Did not score", "Scored")))

nrow(wc)      # 1248 raw
nrow(d)       # 1039 analysis sample
sum(is.na(wc$minutes))

#Optional filtered sample (drops 62 goalkeepers, n = 977)
output <- d %>% filter(pos != "GK") %>% droplevels()

#sample stats
num_vars <- c("age", "minutes", "goals", "assists", "shots", "shots_on_target",
  "fouls", "cards_yellow", "offsides", "interceptions", "tackles_won")
#tables
table1 <- d %>%
  select(all_of(num_vars)) %>%
```

```

pivot_longer(everything(), names_to = "variable", values_to = "value") %>%
group_by(variable) %>%
summarise(n= n(),mean= mean(value),sd= sd(value),min= min(value), Q1= quantile(value,
0.25),median = median(value),Q3= quantile(value, 0.75),max= max(value),skew =
mean((value - mean(value))^3) / sd(value)^3,pct_zero = 100 * mean(value == 0)) %>%
mutate(variable = factor(variable, levels = num_vars)) %>%
arrange(variable) %>%
mutate(across(where(is.numeric), ~round(.x, 3)))
print(table1 n = 20)

```

```

table2 <- d %>%
group_by(pos) %>%
summarise(n= n(),
mean_goals= mean(goals),
mean_shots = mean(shots),
mean_minutes = mean(minutes),
pct_scored = 100 * mean(scored == "Scored")) %>%
mutate(across(where(is.numeric), ~round(.x, 3)))
table2

```

```

n_distinct(d$team_country) # 48 teams
range(table(d$team_country)) # 17 to 25 players per team
table(d$pos)

```

```

#correlations
cor_vars <- c("goals", "shots", "shots_on_target", "assists", "minutes", "age",
" fouls", "offsides", "interceptions", "tackles_won", "cards_yellow")
cormat <- cor(d[, cor_vars])
round(cormat["goals", ], 3)
round(cormat["shots", "shots_on_target"], 3) # 0.886

```

#plots

```

p1 <- ggplot(d, aes(goals)) +
geom_bar(fill = "steelblue") +
labs(title = "(a) Goals scored", x = "goals", y = "count")

```

```

p2 <- ggplot(d, aes(scored, fill = scored)) +
geom_bar(show.legend = FALSE) +
scale_fill_manual(values = c("red", "green")) +
labs(title = "(b) Scored or not", x = NULL, y = "count")

```

```

p3 <- ggplot(d, aes(minutes)) +
geom_histogram(bins = 30, fill = "steelblue") +

```

```

labs(title = "(c) Minutes played", x = "minutes", y = "count")

p4 <- ggplot(d, aes(age)) +
  geom_histogram(binwidth = 1, fill = "steelblue") +
  labs(title = "(d) Age", x = "age (years)", y = "count")

p5 <- ggplot(d, aes(pos, goals)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "(e) Goals by position", x = "position", y = "goals")

p6 <- ggplot(d, aes(shots, goals)) +
  geom_point(colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  labs(title = "(f) Shots vs goals", x = "shots", y = "goals")

p7 <- ggplot(d, aes(minutes, goals, colour = scored)) +
  geom_point() +
  scale_colour_manual(values = c("red", "green")) +
  labs(title = "(g) Minutes vs goals", x = "minutes", y = "goals", colour = NULL)

## ---- Figure 1: the seven panels ----
(p1 | p2 | p3 | p4) / (p5 | p6 | p7)
ggsave("figure1_eda_panels.png", width = 16, height = 8.5, dpi = 300)

#heat map
p_cor <- as.data.frame(as.table(cormat)) %>%
  ggplot(aes(Var1, Var2, fill = Freq)) +
  geom_tile() +
  geom_text(aes(label = sprintf("%.2f", Freq)), size = 3) +
  scale_fill_gradient2(low = "red", mid = "white", high = "blue",
    limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "correlation") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p_cor
ggsave("figure2_correlation_matrix.png", width = 8.5, height = 7.5, dpi = 300)

```